1. //this source file name is main.c

```
#include<stdio.h>
  int sum(int,int);
   int main()
   {
          int x=10,y=20;
        printf("%d",sum(x,y));

    }
```

Note: compile this program step-by-step (gcc -E, -S, -c, etc..)

2. // this file name is sum.c

```
    int sum(int a, int b)
    {
          return a+b;
     }
```

Note: compile this file sum.c using gcc (step by step. Gcc -E, -S, -c etc..)

3.
Compile the above two files using
gcc main.c sum.c

4.
Study nm command , using $man nm
observe the output of $nm a.out

5.
```
   int x=10;
 void inc(void)
 {
    ++x;
 }
 int main()
 {printf("x=%d\n",x);
        inc();
        inc();
        inc();
 printf("x=%d\n",x);}
```

6.
```
  extern int x;
   void  inc(void)
    {
       ++x;
    }
   int main()
   {  printf("x=%d\n",x);
       inc();
       inc();
       inc();
     printf("x=%d\n",x);        }
int x=10;
```

7.
```
    int x;
    void  inc(void)
     {
        ++x;
     }
    int main()
    {
printf("x=%d\n",x);
        inc();
        inc();
        inc();
 printf("x=%d\n",x);
    }
int x=10;
```

8.
```
       static int x=10;
    void  inc(void)
     {
        ++x;
     }
    int main()
    {
printf("x=%d\n",x);
        inc();
        inc();
        inc();
  printf("x=%d\n",x);
    }
```

9.
main.c

```
  int x = 10;
void inc(void);
int main()
{
     inc();
     inc();
     inc();

}
```

inc.c

```
void inc(void)
{
     extern int x;// also try without extern.
     ++x;
}
```

Note: compile two files as
$gcc main.c inc.c

10.
main.c

```
static  int x = 10;
void inc(void);
int main()
{
     inc();
     inc();
     inc();

}
```

inc.c

```
void inc(void)
{
     extern int x;
     ++x;
}
```

Note: compile two files as
$gcc main.c inc.c

11.
```
 int  x=10;
 int main()
{
     int y=x;
      int z=y;
printf("%d %d %d",x,y,z);

}
```

12.
```
int main()
{
    int a=1;
    switch(a)
    {        int b=5;
             case 1: printf("b=%d\n",b);
     }
}
```

13.
```
  //main.c
  int i;
  static void inc(void)
  {
      i++;
   }
  int main()
  {
       inc();
       inc();
     printf("%d", i);
  }
```

14.
```
  //inc.c

        void inc(void)
        {
              int i;
               ++i;
         }
```

Note: compile along with program in 13th. as
  $ gcc main.c inc.c
Also observe the a.out content.
  $ nm a.out